

Deep Learning Project Proposal

Design and Creative Technologies

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ReviewPulse v3: Aspect-Based Sentiment Analysis of Customer Reviews with Attention-Based Deep Learning

1. Abstract

Customer reviews often praise one aspect while criticizing another, yet sentence-level sentiment systems - our Assessment 1 N-gram classifier and the ISY503 ReviewPulse v1.0.0/v2.x review-level classifiers alike - reduce a review to one label. This proposal designs **ReviewPulse v3.0**, a new aspect-based sentiment analysis (ABSA) system that predicts sentiment for a specified aspect rather than the whole review. Four models are compared on SemEval-2014 Restaurants: TF-IDF, a target-agnostic LSTM, ATAE-LSTM and DistilBERT. The study evaluates performance, efficiency and indicative token-level evidence, then proposes a Streamlit v3 prototype for Assessment 3.

2. Problem Statement, Aim and Research Questions

Problem. Most sentiment systems assign one polarity to an entire text. Our Assessment 1 N-gram classifier and the deep CNN of Zhao, Gui and Zhang (2018) do this. However, "*the food was great but the service was slow*" contains opposing opinions about two aspects. A single label hides the detail that product, hospitality and customer-experience teams need to act on.

Aim. Design and evaluate a low-compute ABSA system that predicts sentiment for a specified aspect, measures the value of aspect conditioning, compares ATAE-LSTM with

DistilBERT, and presents indicative token-level evidence where the model supports attention or attribution.

Research questions.

- **RQ1** - How much does explicit aspect conditioning improve sentiment classification on multi-aspect sentences compared with a target-agnostic baseline?
- **RQ2** - How do ATAE-LSTM and DistilBERT compare on accuracy, macro-F1 and computational efficiency?
- **RQ3** - What human-readable evidence do attention or attribution visualizations provide for aspect-level predictions?

3. Literature Review

Deep learning changed sentiment analysis by learning representations rather than relying only on engineered features. Zhao, Gui and Zhang (2018) report that a convolutional network over word vectors outperforms traditional baselines for Twitter sentiment. Nevertheless, their model, like ReviewPulse v1.0.0, produces one label per text. Pontiki et al. (2014) define SemEval-2014 Task 4 and its aspect-level annotations, shifting the practical question from whether a review is positive to which aspect is positive.

Sequence models address this gap through target conditioning. Tang et al. (2016) show that target-dependent LSTMs improve over target-agnostic variants by representing context around the target, although they do not selectively weight every context token. Wang et al. (2016) add an aspect embedding and attention in ATAE-LSTM, allowing the same sentence to be

represented differently for each aspect. This makes ATAE-LSTM a suitable lightweight model and exposes attention weights, but the representation is still learned from a comparatively small benchmark and may overfit.

Transformers offer a different trade-off. Devlin et al. (2019) introduce pretrained bidirectional contextual representations, while Sun, Huang and Qiu (2019) reformulate ABSA as sentence-pair classification using an auxiliary aspect sentence. DistilBERT applies this contextual advantage with lower compute than full BERT, but it remains more expensive than ATAE-LSTM and neither transformer attention nor post-hoc attribution is automatically a faithful causal explanation. Recent transformer benchmarks (Jayakody et al., 2024), zero- and few-shot LLM studies (Simmering & Huoviala, 2023), and the review by Hua et al. (2024) position these models within a rapidly expanding field.

Accordingly, the project does not assume that the Transformer will win. It tests whether pretrained context justifies its additional cost against ATAE-LSTM's lighter, aspect-conditioned design. The comparison therefore combines predictive performance, efficiency and indicative interpretability evidence. Neural aspect discovery (He et al., 2017) remains a possible extension, not a requirement for the core experiment.

4. Proposed Approach and Methods

Dataset and aspect origin. SemEval-2014 Task 4 Restaurants is the primary domain; Laptops is an optional cross-domain extension. Training and evaluation use the dataset's gold aspect terms. The original *conflict* polarity label is removed from the three-class positive/neutral/negative task and its exclusions are counted and reported. In Streamlit v3, a user

instead enters a review and manually specifies one or more aspects. Automatic aspect extraction and Topic Modelling remain outside the core scope.

Model progression. TF-IDF with logistic regression and a target-agnostic LSTM consume only the review and establish sentence-level baselines. ATAE-LSTM consumes *(review, aspect)* through an aspect embedding and aspect-aware attention. DistilBERT also consumes *(review, aspect)*, representing the aspect as an auxiliary sentence. This four-model progression isolates recurrence, explicit conditioning and pretrained contextual representation. As optional exploratory extensions, a GRU variant may probe whether simpler gating changes the sentence-level floor, and a CNN baseline - closer to Zhao, Gui and Zhang (2018) - may test a non-recurrent alternative.

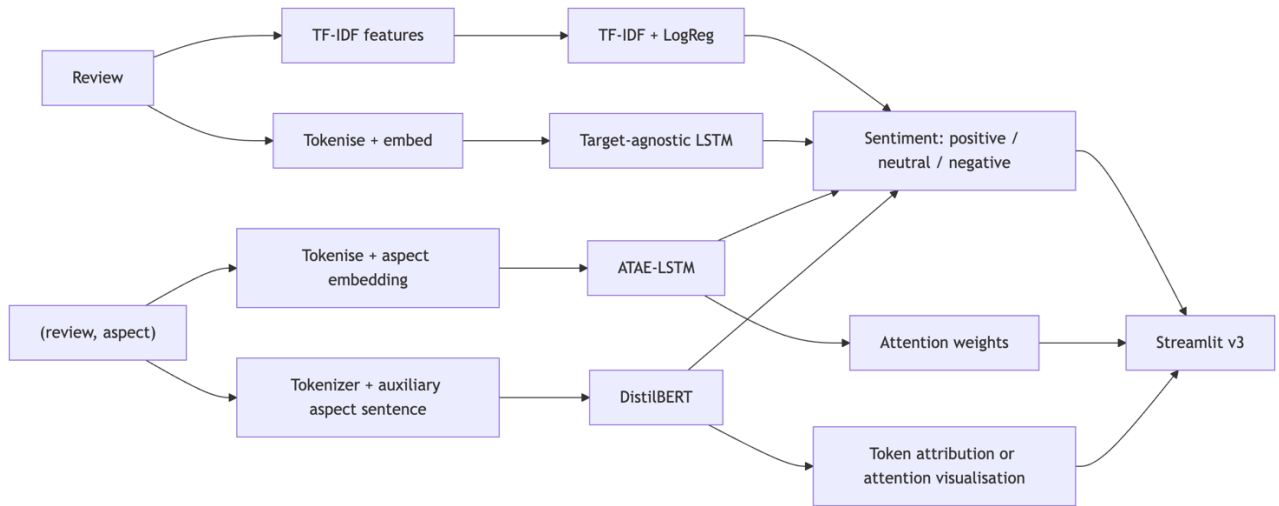


Figure 1. ReviewPulse v3.0 pipeline. The sentence-level baselines receive only the review; the aspect-conditioned models receive a *(review, aspect)* pair. Only attention- or attribution-capable models produce token-level visual evidence.

Evaluation. Accuracy, macro-F1, class-level results and efficiency are reported on fixed train/development/test splits. Splitting is grouped by sentence before aspect instances are formed, preventing the same review from leaking across partitions. A **mixed-polarity multi-aspect subset** contains sentences with at least two aspects associated with different polarities; it tests the cases that one sentence-level label cannot resolve.

Table 1. Assessment 3 reporting shell. The last column evaluates sentences with at least two aspects carrying different polarities.

Model	Accuracy	Macro-F1	Accuracy on mixed-polarity multi-aspect subset
TF-IDF + logistic regression (review only)	to be reported (A3)	to be reported (A3)	to be reported (A3)
Target-agnostic LSTM (review only)	to be reported (A3)	to be reported (A3)	to be reported (A3)
ATAE-LSTM (review, aspect; attention)	to be reported (A3)	to be reported (A3)	to be reported (A3)
DistilBERT (review, aspect; auxiliary sentence)	to be reported (A3)	to be reported (A3)	to be reported (A3)

Interpretability output. For predictions from attention- or attribution-capable models, the interface presents the aspect label and indicative token-level evidence. ATAE-LSTM exposes attention weights; DistilBERT uses token attribution or an attention visualization. Table 2 is an illustrative expected interface output, not an experimental result. TF-IDF and the target-agnostic LSTM are not required to produce heatmaps.

Table 2. Illustrative expected output for one multi-aspect review. Bold text stands in for token shading.

Attention is indicative evidence, not a causal explanation unless faithfulness tests support that interpretation.

Review: "The food was great but the service was slow."

Aspect	Predicted sentiment	Indicative token-level evidence (bold = highest weight)
food	positive	the · food · was · great · but · service · was · slow
service	negative	the · food · was · great · but · service · was · slow

Deployment. The Assessment 3 prototype lets a user enter a review plus one or more aspects, then returns a sentiment for each aspect and model-supported visual evidence.

5. Project Plan and Risk Management

Work is divided into dated, reviewable outcomes (Table 3). Luis owns the technical foundation; group ownership marks integration points requiring peer review and shared acceptance. A fixed seed, versioned data audit, saved configurations and a single evaluation script support reproducibility. The group will review milestone evidence at each hand-off and reduce scope before moving a due date. Appendix B records probability, impact, mitigation and a concrete contingency for technical and collaboration risks.

Table 3. Delivery plan from data audit to Assessment 3 package.

Phase	Date	Outcome	Owner
Data + baseline	18-26 Jul	Audited dataset and TF-IDF baseline	Luis
LSTM + ATAE-LSTM	27 Jul – 2 Aug	Checkpoint and metrics	Luis
DistilBERT	3-8 Aug	Trained transformer	Victor
Evaluation + interface	9-13 Aug	Comparison and Streamlit v3	Juan
Report + package	14-19 Aug	Report, code and ZIP	Group

The critical path is dataset audit, aspect-safe splitting, model training, common evaluation, interface integration and packaging. Zero-cost compute constrains Transformer experiments, so the accepted minimum product remains the audited baselines, ATAE-LSTM, shared evaluation and working interface. DistilBERT is retained when compute and validation checks pass; optional Laptops transfer, automatic aspect extraction, Topic Modelling, a GRU variant and a CNN baseline are cut first.

6. Conclusion

Sentence-level sentiment loses actionable aspect detail. ReviewPulse v3.0 proposes a controlled four-model comparison, with Restaurants as the core domain, explicit aspect-safe evaluation and a dated delivery plan. The prototype will visualize indicative token-level evidence where supported, without presenting attention as model reasoning or a causal explanation.

7. Appendices

7.1. Appendix A – From Assessment 1 to ReviewPulse v3.0

This appendix separates two lineages that are easy to conflate: the DLE602 assessment sequence (Assessment 1 to this proposal) and the ReviewPulse product releases (built independently for ISY503). The numbered list states plainly what ReviewPulse v3.0 does and does not inherit from the earlier releases; Table A1 then tracks the narrower Assessment 1 to Assessment 2/3 transition.

1. **DLE602 Assessment 1 - N-gram sentiment classifier.** A conceptual precursor only: a transparent, hand-built bigram model over Twitter data. It is not part of the ReviewPulse codebase and was never released as ReviewPulse v1.0.0.
2. **ReviewPulse v1.0.0 (ISY503, 2026-04-26).** The first ReviewPulse release: a TF-IDF + Logistic Regression baseline and a BiLSTM + GloVe neural model, both predicting one binary sentiment label (positive/negative) per whole Amazon review across four product domains.
3. **ReviewPulse v2.x (ISY503, 2026-04-29 onward).** Adds a fine-tuned DistilBERT transformer alongside the v1.0.0 models, plus a modular package refactor, expanded test coverage and a hardened Streamlit deployment. The prediction unit is unchanged: one label per whole review.
4. **ReviewPulse v3.0 (DLE602, this proposal).** A new implementation, not an extension of the same codebase's results: SemEval-2014 Restaurants replaces the Amazon dataset, the label space becomes three-class aspect sentiment (positive/neutral/negative), and the prediction unit becomes one label per

(review, aspect) pair. All four models are trained from scratch on this new task; no ReviewPulse v1.0.0/v2.x checkpoint, metric or result is reused as an Assessment 3 outcome.

ReviewPulse v3.0 reuses the ISY503 project's engineering discipline: the preprocessing/training/evaluation package structure, the fixed-seed and versioned-artifact conventions, and the Streamlit deployment pattern. It reuses none of the ISY503 results: the task, dataset, label space, model inputs and evaluation protocol described in Sections 4-5 are all new for this proposal, and the ABSA system remains a design, not a completed build - Table 1 and Table 2 report placeholders, not measured outcomes.

Table A1 – Knowledge transition from Assessment 1 to the proposed Assessment 2 and 3 project.

Assessment 1 - N-gram sentiment	Assessment 2/3 - ReviewPulse v3.0
Count observed word sequences	Learn distributed and contextual representations
Fixed Markov context window	LSTM recurrence, aspect-aware attention and Transformer context
One sentiment label per tweet	One sentiment label per aspect within a review
Add-k smoothing controls sparse counts	Dropout, weight decay, early stopping and transfer learning control overfitting
Hand-defined probability and threshold rule	Learned logits optimized through loss and backpropagation
Inspect bigram probabilities and error examples	Inspect attention or attribution alongside error examples
Accuracy, macro-F1 and confusion matrices	Retain the same metrics, now calculated for aspect-level predictions

ReviewPulse v1.0.0 and v2.x also supply reusable preprocessing, experiment, metric and interface patterns. The Assessment 3 implementation changes the model input from review-

only baselines to explicit (*review, aspect*) pairs for ATAE-LSTM and DistilBERT, then exposes per-aspect predictions in Streamlit v3.

7.2 Appendix B – Risk Register

Table B1 – Risk Register and Mitigation plan.

Risk	Probability	Impact	Mitigation	Contingency
Overfitting on the small benchmark	High	High	Stratified development checks, dropout, early stopping and fixed seeds	Reduce model capacity and report variance across seeds
Transformer compute exceeds the free tier	Medium	High	DistilBERT, short pilot runs, capped sequence length and checkpointing	Ship ATAE-LSTM as the primary deep model and report the compute constraint
Leakage between aspects from the same sentence	Medium	High	Split by sentence identifier before expanding aspect instances; audit overlaps	Rebuild all splits and rerun every affected result
Artifact or model loading failure	Medium	High	Pin dependencies, save configs and test a clean-start load before packaging	Bundle a verified lightweight checkpoint and fall back to CPU inference
Scope creep from Laptops, extraction or Topic Modelling	High	Medium	Gate all extensions until the core acceptance tests pass	Remove extensions in that order and preserve the Restaurants core
Unequal group contribution	Medium	High	Named owners, dated evidence, twice-weekly reviews and a contribution log	Reassign overdue work within 24 hours and document the revised ownership

End of Appendix Section

Statement of Acknowledgment

We acknowledge that we have used the following AI tools in the preparation of this project proposal:

- OpenAI ChatGPT (Codex-5.5)
- Anthropic Claude (Opus 4.8)

Both tools were used to assist with clarifying aspect-based sentiment analysis (ABSA) concepts and the progression from N-gram baselines to attention-LSTM and transformer models, structuring the literature review and its critical synthesis, sanity-checking the SemEval-2014 experimental design against data leakage, refining academic clarity, and supporting APA 7th referencing conventions. All source papers were read and verified against the original publications by the group.

Prompt examples:

1. *"Given our Assessment 1 N-gram single-label classifier, how do I frame the aspect-level problem so that a sentence like 'the food was great but the service was slow' motivates ABSA over sentence-level sentiment?"*
2. *"Compare ATAE-LSTM (Wang et al., 2016) and BERT-for-ABSA via auxiliary sentence (Sun et al., 2019) as two contrasting models on SemEval-2014 Task 4, and explain what each adds over a target-agnostic baseline."*
3. *"Review this project plan for a hypothetical ABSA proposal: are the risks, contingency plan and critical success factors sound and feasible within a fixed academic timeframe and a zero-cost compute budget?"*

We confirm that the use of these AI tools has been in accordance with the Torrens University Australia Academic Integrity Policy and the TUA, Think and MDS Position Paper on the Use of AI. We confirm that the final output is authored by the group and represents our own critical thinking, analysis, and synthesis of sources. We take full responsibility for the final content of this report.

8. References

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